

Preclass Learning: Introduction to Data Science (ICA 1)

1. The Data Science Pipeline

Data science is often described as a cyclical, iterative process that converts raw data into actionable knowledge:

1. **Ask:** Formulating a clear, answerable question.
2. **Acquire/Import:** Getting the data into the computing environment.
3. **Wrangle/Tidy:** Cleaning, transforming, and structuring the data.
4. **Explore:** Visualizing and summarizing the data to find patterns and anomalies.
5. **Model/Analyze:** Applying statistical or machine learning models to answer the question.
6. **Communicate:** Presenting the findings, results, and limitations to an audience in a clear, compelling manner.

2. Key Disciplinary Components

Data science sits at the intersection of three major areas:

- **Statistics:** Provides the mathematical rigor, methods for inference, and tools for quantifying uncertainty.
- **Computer Science:** Contributes the programming skills, algorithmic efficiency, and knowledge of data structures.
- **Domain Expertise:** Subject-matter knowledge needed to correctly interpret the data, vet assumptions, and ensure the questions being asked are relevant.

3. Literate Programming

The concept of combining code, output, and narrative text into a single, reproducible document. Tools like **Quarto** or R Markdown facilitate this, ensuring that the results presented can be instantly verified by running the code.